

Interaction Structure and Initiative Control in AI-Assisted Decision Support Systems

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Abstract—This paper presents the design and model validation of a controlled behavioral experiment investigating how the *timing of AI advice* affects decision quality in human–AI decision support systems (DSS). While predictive models are typically evaluated using offline metrics such as accuracy and calibration, real-world outcomes depend on how human decision-makers interact with model outputs.

We construct a controlled experimental DSS using a calibrated logistic regression model for credit default prediction ($AUC = 0.691$, $ECE = 0.033$) and a model-independent notion of *case difficulty* derived from empirical base rates over discretized feature space. Holding the predictive model fixed, we design a within-subject experiment that manipulates the interaction protocol (AI-first, human-first, no-AI) and measures how these protocols affect reliance behavior and the translation of model-level performance into system-level outcomes.

We additionally introduce a normative decision benchmark based on asymmetric decision costs to evaluate alignment between human–AI decisions and cost-optimal policies. Preliminary model validation results confirm that the experimental infrastructure meets the requirements for a non-degenerate reliance environment. Data collection targeting $N = 100$ participants is planned for immediate deployment.

Index Terms—Human-AI Interaction, Decision Support Systems, Interaction Timing, Reliance, Calibration, Decision Theory.

I. INTRODUCTION

Predictive models are increasingly deployed to assist human decision-makers in high-stakes domains: credit assessment, clinical triage, and recidivism prediction. The standard evaluation pipeline treats model performance as a proxy for deployment quality. It is not. A model’s predictions influence outcomes only insofar as decision-makers choose to act on them, and identical models can produce substantially different real-world outcomes depending on how their recommendations are presented [1], [2].

This gap between model-level and system-level performance is well-documented. Kleinberg et al. [1] showed through retrospective policy simulation that an algorithm substantially outperformed judges at predicting flight risk, yet the potential gains depend entirely on how such predictions would be integrated into actual decisions. Green and Chen [2] tested this integration directly in a behavioral experiment and found that while algorithmic risk scores reliably improved participant accuracy over unaided baselines, participants failed to match the

model’s standalone performance, could not evaluate prediction quality, and exhibited racial bias in their interactions with the AI. A consistent conclusion is that the integration mechanism, not model quality, determines realized performance.

A central but underexplored dimension of this integration is *interaction timing*: whether AI advice is presented before or after the human forms an initial judgment. AI-first protocols risk anchoring the human’s reasoning to the model’s output regardless of its correctness [8]. Human-first protocols preserve independent deliberation but may reduce the magnitude of adjustment toward AI advice [9]. The empirical question of which ordering produces better-calibrated decisions, and under what conditions, remains open.

We address this through a controlled behavioral experiment using a calibrated logistic regression model for loan default prediction. Holding the model fixed across all conditions, we manipulate only the interaction protocol (AI-first, human-first, no-AI) and measure effects on decision accuracy, reliance behavior, and alignment with cost-optimal decision rules.

This paper makes three specific contributions. First, we introduce a *model-independent* operationalization of case difficulty based on Bayesian-smoothed empirical base rates, which avoids the circularity present in studies that define difficulty relative to the model being evaluated. Second, we implement a within-subject Latin square design that isolates the causal effect of interaction timing while holding the predictive model, case set, and difficulty distribution fixed. Third, we propose a *normative cost-sensitive benchmark* that evaluates human–AI decision quality against optimal behavior under asymmetric loss, separating decision quality from both raw accuracy and model agreement.

II. LITERATURE REVIEW

A. The Model-System Performance Gap

A foundational observation motivating this work is that deploying an accurate predictive model does not automatically improve decisions. Kleinberg et al. [1] demonstrated this through a retrospective policy simulation of bail decisions, showing via counterfactual analysis that an algorithm substantially outperformed judges at predicting flight risk, with potential welfare gains including crime reductions of up to 24.7% at fixed jailing rates. Their analysis was conducted

on historical data rather than in a live human-AI interaction setting, leaving open the question of whether such gains would materialize when decision-makers actually use algorithmic predictions.

Green and Chen [2] addressed this question directly through a controlled behavioral experiment. They found that algorithmic risk scores reliably improved participant accuracy over unaided baselines in most conditions, but that participants failed to match the model’s standalone accuracy. Critically, participants could not effectively evaluate prediction quality, did not calibrate their reliance based on the model’s actual performance, and exhibited racial bias in their interactions with the risk assessment. We adopt their “algorithm-in-the-loop” framework as our central lens, treating the gap between model accuracy and realized decision quality as the primary quantity of interest.

Mozannar and Sontag [6] approached this gap from a machine learning perspective, formalizing the *learning to defer* problem: given that a human expert will make the final decision, how should the model be trained to allocate decision authority optimally? Their framework treats human performance as a function of case difficulty and formalizes the system-level objective that this paper studies empirically. We complement their theoretical contribution with behavioral data on how interaction structure, rather than model architecture, determines whether the potential for complementary performance is realized.

B. Reliance, Overreliance, and Automation Bias

The human factors literature has long documented the tendency toward *automation bias*: the uncritical acceptance of automated recommendations even when human judgment would be superior [4]. In AI-assisted decision settings, this manifests as overreliance (following AI advice when the model is incorrect), which is systematically documented across domains including clinical diagnosis, recidivism prediction, and financial assessment [3], [7].

Buçinca et al. [3] showed that *cognitive forcing functions*, design interventions that require users to form an independent judgment before receiving AI advice, can substantially reduce overreliance. Their finding is directly relevant to our human-first protocol, which embeds a structural cognitive forcing mechanism by collecting an initial decision before revealing the AI prediction. Their work, however, does not systematically vary the timing of AI advice across a common case set while controlling for model quality and case difficulty, which is the precise design we implement.

Schemmer et al. [7] proposed the *Appropriateness of Reliance* (AoR) framework, decomposing reliance into two dimensions: correct AI reliance (following the AI when it is correct) and correct self-reliance (overriding the AI when it is wrong). This two-dimensional view is essential: a participant who always follows the AI scores high on the first dimension but near zero on the second. We operationalize a related decomposition in our reliance analysis, distinguishing bene-

ficial reliance, harmful override, over-reliance, and beneficial override across protocol conditions.

C. Interaction Timing and Anchoring

The role of cognitive biases in AI-assisted decision making has received growing empirical attention. Rastogi et al. [8] mathematically modeled anchoring bias in the human-AI collaboration setting and proposed a time allocation strategy to mitigate it. Their key finding was that forcing participants to spend more time deliberating (through enforced time delays before submission) effectively reduced anchoring on AI predictions and improved collaborative performance, particularly when the AI model had low confidence and was incorrect. While their work addresses de-anchoring through time pressure manipulation rather than protocol ordering, their theoretical framework for quantifying anchoring bias informs our measurement approach.

Steyvers and Kumar [9] identify interaction timing as one of three unresolved challenges in AI-assisted decision making, distinguishing between concurrent, sequential, and on-demand paradigms. They note that the time-delayed paradigm, where a forced waiting period is imposed before the human can submit, has shown promise for reducing anchoring on AI advice. Our design extends this line of inquiry by manipulating not just the timing of the decision but the *ordering* of information: whether the human forms a judgment before or after seeing AI advice. The Weight of Advice metric [13], computed in our human-first condition where both initial and final estimates are collected, provides a direct measure of how much participants adjust toward the AI prediction.

D. Confidence, Calibration, and Uncertainty Communication

How AI uncertainty is communicated affects not only reliance behavior but participant calibration. Zhang et al. [10] demonstrated that displaying model confidence alongside predictions can improve trust calibration, meaning participants become more discriminating about when to follow AI advice. However, this benefit depends on the confidence signal being genuinely informative; when confidence is poorly calibrated, displaying it can increase overreliance.

This motivates our decision to use a probability-based AI representation rather than a binary recommendation alone. Our system displays both a predicted default probability and a derived binary recommendation, allowing us to distinguish compliance with the recommendation (binary matching) from probability anchoring (continuous adjustment). The model’s predictive certainty (the probability assigned to the predicted class) averages 0.817 when correct and 0.731 when incorrect, yielding a gap of 0.086. This gap is large enough for the certainty signal to be informative but small enough to create a genuinely non-degenerate reliance environment.

E. Case Difficulty and Complementary Performance

The relationship between task difficulty and human-AI team performance is a central concern for achieving complementary

outcomes, where the team exceeds either member’s solo accuracy. Bansal et al. [5] investigated whether AI explanations could help achieve such complementary performance. They found that while AI augmentation (providing predictions) did produce complementary improvements, explanations did not increase these gains. Instead, explanations increased the probability that participants would accept the AI’s recommendation regardless of its correctness. A key implication is that achieving complementary performance requires the human to selectively override incorrect AI predictions, which demands that human and AI errors be non-overlapping.

Our experimental design is structured to test this directly. We construct 18 cases across three difficulty tiers (easy, medium, hard) and two model correctness conditions, with each block containing cases where the model is both correct and incorrect. This creates opportunities for both beneficial reliance (following correct AI) and beneficial override (correcting incorrect AI), enabling us to measure whether interaction timing affects the balance between these behaviors. The medium-difficulty tier is of particular interest: these cases fall closest to the cost-sensitive decision threshold $\tau = 0.167$, where model-optimal disagreement is highest (18.7% model-optimal agreement versus 37.6% for hard cases), creating the richest decision-theoretic environment for studying human-AI alignment.

F. Research Gap

The existing literature on human-AI decision support addresses reliance behavior, explanation design, confidence communication, and cognitive forcing functions largely in isolation. What remains underspecified is how *interaction structure*, specifically the ordering of human judgment and AI advice, modulates the translation of model-level performance into system-level outcomes across cases that vary in genuine difficulty.

Most prior work either treats timing as a secondary variable, studies it in single-condition designs that cannot isolate protocol effects, or evaluates it without a normative benchmark that separates model performance from decision optimality. This paper addresses all three gaps: timing is the primary independent variable, held against a fixed model; the within-subject design enables direct comparison across protocols; and a cost-sensitive normative benchmark allows evaluation of decision quality beyond raw accuracy.

III. METHODOLOGY

A. Hypotheses

- **H1 (AI Benefit):** AI assistance improves final decision accuracy relative to unaided human decisions.
- **H2 (Difficulty Moderation):** The benefit of AI assistance varies across difficulty levels, with different reliance patterns in easy, medium, and hard cases.
- **H3 (Timing Effect):** The timing of AI advice (AI-first vs. human-first) significantly affects the degree of probability anchoring, measured as the absolute distance between

participant and model probability estimates, and reliance behavior.

- **H4 (Selective Reliance):** Participants rely more heavily on AI predictions when the model is correct than when it is incorrect, as measured by Weight of Advice.
- **H5 (Protocol and Calibration):** Participants in the human-first condition exhibit better probability calibration (lower ECE) than those in the AI-first condition, because forming an independent judgment before receiving AI advice reduces anchoring on model predictive certainty.
- **H6 (Normative Deviation):** Participants systematically deviate from cost-sensitive optimal decisions, with the largest deviations in medium-difficulty cases, where the model-optimal disagreement is highest.

B. Data and Feature Engineering

The dataset consists of approximately 1.3 million LendingClub loans issued between 2007 and 2018, with a binary default prediction task and an observed default rate of approximately 20%. Only features available at loan origination are used to prevent data leakage. Post-outcome variables, payment history fields, and internal credit grading are removed prior to modeling. A temporal split is applied: training data includes loans issued before 2016; test data covers loans from 2016 onward. A distribution shift is observed between periods, with the default rate increasing from 18.4% to 22.9%, reflecting changing macroeconomic conditions. This shift is intentionally preserved, as it reflects realistic decision-making conditions.

The final feature set includes loan amount, term, interest rate, log-transformed annual income, debt-to-income ratio, revolving utilization, home ownership, loan purpose, and credit history length. Feature selection prioritizes interpretability and pre-origination availability over predictive accuracy, consistent with the goal of a cognitively tractable decision environment. Redundant features (e.g., installment, which is functionally derived from loan amount, term, and rate; $\rho = 0.95$) and near-zero-signal features are excluded based on univariate analysis and Spearman rank correlations.

Two representations are maintained in parallel: continuous features are used for model training, while human-readable labels derived from training-set quantile bins are used in the experiment interface. This separation ensures the model operates on full-resolution data while participants receive interpretable, discretized information.

Case difficulty is defined independently of model predictions using Bayesian-smoothed empirical default rates computed over discretized feature bins (interest rate, DTI, and income) on training data only. The smoothing parameter $\alpha = 50$ shrinks bin-level estimates toward the global default rate, reducing noise in sparse bins. Difficulty is operationalized as the distance of the bin-level base rate from maximum uncertainty (0.5), yielding a score in $[0, 1]$ where 1 represents maximum ambiguity. Fixed thresholds partition cases into easy (≤ 0.33), medium (0.33–0.66), and hard (> 0.66) tiers. This construction avoids circularity between model predictions

and difficulty assignment, allowing difficulty to serve as an exogenous variable.

C. Predictive Model

A logistic regression model with isotonic calibration is used as the DSS engine. The training pipeline employs a four-way temporal split: training set (chronological 70% of pre-2016 data), calibration set (first half of remaining pre-2016 data), validation set (second half), and test set (post-2016). This isolation prevents the calibration procedure from memorizing its fitting data, which would produce artificially low ECE estimates.

Three candidate models were evaluated (logistic regression, XGBoost, histogram gradient boosting). Bootstrap stability analysis over 10 resampled training sets confirmed high consistency across models (Brier score standard deviation < 0.001 for all three). Paired t -tests on bootstrap Brier scores showed statistically significant but practically negligible differences ($\Delta_{\text{Brier}} \approx 0.001$). Logistic regression was selected for its transparency, which minimizes confounding factors attributable to model complexity rather than interaction structure. The deliberate use of a moderate-performing model (AUC = 0.691) reflects the research goal: the study investigates how humans integrate imperfect AI advice, and a model with moderate accuracy creates the conditions necessary for reliance decisions to be non-trivial. Section IV reports detailed model performance.

D. Experimental Design

A within-subject Latin square design is used. Eighteen cases are pre-selected from the test set using stratified sampling across difficulty tier (easy, medium, hard) and model correctness (correct, incorrect at the 0.5 classification threshold), yielding six cells of three cases each. Model correctness for case stratification is defined at the standard 0.5 classification threshold and is independent of the cost-sensitive recommendation threshold τ displayed to participants. Cases are organized into three blocks of six, each containing two cases per difficulty level and three per correctness condition. Protocol assignment follows a Latin square rotation across three participant groups, ensuring that every case is evaluated under every protocol across the full sample, and that the protocol-by-difficulty interaction is identifiable within-subject.

TABLE I
LATIN SQUARE PROTOCOL ROTATION

Group	Block 1	Block 2	Block 3
Group 1 ($n \approx 33$)	No-AI	Human-first	AI-first
Group 2 ($n \approx 33$)	Human-first	AI-first	No-AI
Group 3 ($n \approx 34$)	AI-first	No-AI	Human-first

The three protocols are:

- **No-AI:** Participant evaluates case features and makes a decision without model input.

- **AI-first:** The AI assessment (predicted default probability and cost-sensitive binary recommendation) is displayed before the participant makes any decision.
- **Human-first:** Participant makes and submits an initial decision; the AI assessment is then revealed; participant may revise.

Within each block, trial order is randomized per participant to prevent sequence effects. Two practice trials with extreme-probability cases are presented before Block 1 to familiarize participants with the interface and anchor the perceived range of AI outputs; these are excluded from analysis.

Note that the 18 selected cases have a true default rate of 50% (by construction, from balanced correctness stratification), which differs from the population default rate of approximately 23%. Participants are not informed of case-level base rates, and this design choice is necessary to ensure sufficient statistical power for comparing reliance behavior across model-correct and model-incorrect conditions. This is consistent with standard practice in human-AI experimental design [5].

E. Data Quality and Exclusion Criteria

To ensure data quality, the experiment includes a pre-experiment comprehension quiz (verifying understanding of the decision task, the meaning of default probability, and the interpretation of AI advice), attention-check trials with extreme cases embedded in the practice sequence, and time-based exclusion criteria for trials completed below a minimum threshold. Participants who fail the comprehension quiz, exhibit zero response variance across all 18 trials, or complete more than one-third of trials below the time threshold will be excluded from analysis. These criteria are specified before data collection begins.

F. Participants and Recruitment

We plan to recruit approximately $N = 100$ participants from a general-population sample. This is consistent with the majority of prior work on AI-assisted decision making [2], [3], [5], which similarly uses non-expert participants. The contribution of this study concerns the structural effect of interaction timing on reliance behavior, not the modeling of expert loan officer decision processes. All participants receive a standardized briefing on the task and the asymmetric cost structure. The study involves no deception, collects no personally identifiable information, and presents no risk beyond routine cognitive tasks. Participants provide informed consent before beginning the experiment.

G. Sample Size Justification

We conducted simulation-based power analyses with 500 replications per sample size under conservative effect size assumptions derived from prior work [2], [12]. For H1 (AI benefit), tested on binary decision accuracy via a within-subject paired comparison, $N = 100$ yields power > 0.90 to detect a 7-percentage-point accuracy improvement (OR ≈ 1.4). For H3 (timing effect), power depends critically on

the outcome measure. A simulation-based analysis showed that testing H3 on binary accuracy yields power of only ~ 0.25 at $N = 100$, because binary outcomes discard the continuous probability information collected on each trial. Testing H3 on the mean absolute distance between participant and model probability estimates, a continuous measure that captures the degree of anchoring, yields substantially higher power (> 0.80 at $N = 100$ for a 4-percentage-point mean difference, $d \approx 0.3$). Accordingly, the primary test of the timing effect uses continuous probability adjustment, with binary accuracy as a secondary analysis. Two-way interactions are estimable with reduced power; the three-way interaction is treated as exploratory.

H. Trial Structure and Measurement

Each trial presents a standardized loan application profile with human-readable feature labels (e.g., income categorized as LOW, MEDIUM, or HIGH based on training-set quintiles). When the protocol includes AI advice, participants see the model’s predicted default probability as a percentage and a derived binary recommendation (Approve or Reject), computed at the cost-sensitive threshold $\tau = 0.167$. Displaying both the probability and the recommendation allows us to distinguish probability anchoring from compliance with the binary decision, and to infer the participant’s implicit decision threshold.

Each trial collects:

- 1) A binary decision (Approve or Reject)
- 2) A probability estimate of default (0–100 slider, recorded as $[0, 1]$)
- 3) A self-reported confidence rating (0–100 slider)

In the human-first condition, items (1) and (2) are collected twice: once before and once after AI advice is revealed. A threshold-consistency flag is computed post-hoc to identify trials where the binary decision and probability estimate are logically inconsistent. Inconsistency is evaluated at two thresholds: the standard classification boundary (0.5) and the cost-sensitive threshold ($\tau = 0.167$), since a participant who estimates $P(\text{default}) = 0.3$ and rejects may be acting rationally under cost-asymmetry.

This hybrid collection scheme enables computation of Weight of Advice on continuous probability values (human-first condition only), Expected Calibration Error across all conditions, and binary decision-level metrics (accuracy, reliance type) universally.

I. Normative Decision Benchmark

To evaluate alignment between human-AI decisions and decision-theoretically optimal behavior, we define a cost-sensitive decision rule based on asymmetric loss. Let C_{FN} denote the cost of approving a defaulting loan (missed default) and C_{FP} the cost of rejecting a creditworthy applicant (missed opportunity). The optimal threshold is:

$$\tau = \frac{C_{FP}}{C_{FP} + C_{FN}} \quad (1)$$

We set $C_{FN} : C_{FP} = 5 : 1$, yielding $\tau \approx 0.167$. This ratio reflects the well-documented asymmetry in consumer lending, where losses from default typically exceed foregone interest income by a factor of 4–6 $\times$ [11]. A sensitivity analysis across $C_{FN} \in [3, 7]$ confirms that the qualitative pattern of model-optimal disagreement by difficulty tier is robust to the specific cost parameterization. At $\tau = 0.167$, model-optimal disagreement is highest in medium-difficulty cases (18.7% agreement versus 37.6% for hard cases and 79.9% for easy cases), making the medium tier the most decision-theoretically consequential.

J. Statistical Analysis

Hypotheses are evaluated using a tiered analysis strategy informed by the power analysis. H1 (AI benefit on binary accuracy) and H3 (timing effect on continuous probability distance) are the primary confirmatory analyses. H1 is tested via the GLMM specified in Appendix B. H3 is tested primarily using a linear mixed-effects model on the absolute distance between participant and model probability estimates, with binary accuracy as a secondary outcome. H2 (difficulty moderation) and H4 (selective reliance via WOA) are tested as planned comparisons. H5 (calibration) is evaluated by comparing participant-level ECE across protocol conditions. H6 (normative deviation) is evaluated by comparing deviation rates across difficulty tiers. Exploratory analyses include decision revision rates, over-reliance and beneficial override rates, block-order effects to assess carryover, and the Discrimination Ratio defined in Appendix A.

IV. PRELIMINARY MODEL VALIDATION

This section reports results from the model development and case design phases, establishing that the experimental infrastructure meets the requirements for a valid behavioral study. No participant data have been collected at this stage.

A. Model Performance

Table II summarizes model performance on the held-out test set (post-2016, $n = 479,744$). All three models achieve comparable predictive performance, with Brier score differences on the order of 0.001.

TABLE II
MODEL COMPARISON ON TEST SET

Model	AUC	Brier	ECE	Calibration
Logistic	0.691	0.164	0.033	Isotonic
XGBoost	0.697	0.163	0.034	Isotonic
HGB	0.697	0.163	0.033	Sigmoid

Bootstrap stability analysis over 10 resampled training sets confirmed that all models show Brier score standard deviations below 0.001. Paired t -tests on bootstrap Brier scores yielded $\Delta_{\text{Logistic-HGB}} = 0.0012$ ($p < 0.001$) and $\Delta_{\text{Logistic-XGB}} = 0.0011$ ($p < 0.001$). These differences, while statistically significant due to the large test set, are practically negligible. Logistic regression was selected as the final model for its interpretability and transparency.

The model maintains calibration despite a distribution shift between training and test periods: the default rate increased from 18.4% to 22.9%, yet ECE remains at 0.033 after isotonic recalibration. This robustness under distributional shift is itself a meaningful finding, as it confirms that the experimental DSS will present participants with well-calibrated probability estimates even under realistic deployment conditions.

B. Calibration Quality

The selected model achieves $ECE = 0.033$ on the test set after isotonic calibration. The mean predicted probability of default ($\bar{p} = 0.215$) is close to but slightly below the observed test default rate (0.229), reflecting a mild calibration gap attributable to the distribution shift.

The model’s predictive certainty (the probability assigned to the predicted class) averages 0.817 when correct and 0.731 when incorrect, yielding a gap of 0.086. This gap is large enough for the certainty signal to be informative but small enough that reliance decisions are non-trivial.

C. Case Design Validation

TABLE III
MODEL PERFORMANCE BY DIFFICULTY TIER (FULL TEST SET)

Tier	n	Accuracy	Avg. Pred.	Opt. Agree
Easy	241,081	0.863	0.113	0.799
Medium	175,439	0.701	0.276	0.187
Hard	63,224	0.589	0.435	0.376

Table III confirms that the difficulty construction produces monotonically decreasing model accuracy across tiers and that medium-difficulty cases show the lowest normative agreement rate (18.7%), validating the design choice to focus on this tier for the strongest test of H6. The 18 selected experimental cases are balanced across all design cells: 6 per difficulty tier, 9 model-correct and 9 model-incorrect, 6 per block, with each within-block difficulty $\times$ correctness cell containing exactly one case.

The predicted probabilities for the 18 cases range from 0.055 to 0.423 (mean 0.205), reflecting the realistic distribution of credit risk model outputs where the majority of predictions fall in the low-to-moderate range. This compressed probability range is a feature of the lending domain rather than a limitation: the decision-theoretic tension arises not from extreme probabilities but from the gap between the standard classification threshold (0.5) and the cost-sensitive threshold ($\tau = 0.167$), within which most cases fall.

V. PLANNED EVALUATION AND EXPECTED CONTRIBUTIONS

A. Primary Analyses

The planned experimental evaluation will proceed along three axes. First, we will test whether AI assistance improves binary decision accuracy relative to the no-AI baseline (H1), evaluated via the GLMM specified in Appendix B. Second, we will test whether the timing of advice modulates the degree

of probability anchoring (H3), using the absolute distance between participant and model probability estimates as the primary continuous outcome. This measure captures whether AI-first participants anchor more tightly to the model’s prediction than human-first participants, and is substantially more powerful than binary accuracy for detecting timing effects (simulation analysis indicates power > 0.80 at $N = 100$ for the continuous measure versus ~ 0.25 for binary accuracy).

Third, we will examine the reliance decomposition (beneficial reliance, harmful override, over-reliance, beneficial override) across protocols and difficulty tiers. The Discrimination Ratio ($DR > 1$ indicating selective override of incorrect AI) is the key indicator of appropriate reliance. The human-first condition, which embeds a cognitive forcing function, is predicted to yield higher DR than the AI-first condition.

The normative benchmark analysis will quantify the gap between participant decisions and cost-optimal behavior at $\tau = 0.167$, with the strongest test in the medium-difficulty tier where model-optimal disagreement is largest in the population.

B. Exploratory Analyses

Exploratory analyses include Weight of Advice distributions (human-first only), block-order effects to assess learning and fatigue across the within-subject sequence, participant-level threshold estimation from the relationship between probability estimates and binary decisions, and the confidence-accuracy correlation as a measure of metacognitive calibration.

C. Expected Contribution

If the hypotheses are supported, this study will provide empirical evidence that interaction timing is a first-order determinant of human-AI decision quality, one that operates independently of model accuracy, calibration, or explanation quality. The model-independent difficulty construction enables generalization of the timing effect across difficulty conditions without the circularity introduced by model-dependent difficulty definitions. The normative benchmark provides a principled evaluation criterion that goes beyond accuracy to measure alignment with optimal behavior under decision-theoretic constraints.

The practical implication is that the same predictive model can be made more or less effective through interaction design alone, which has direct consequences for the deployment of AI-assisted decision systems in domains where human oversight is required.

VI. DISCUSSION AND CONCLUSION

This paper presents the complete design of a controlled behavioral experiment studying how interaction structure affects the translation of model-level performance into system-level outcomes in AI-assisted decision making. The model validation results establish that the experimental infrastructure, including a calibrated logistic regression model, a model-independent difficulty taxonomy, and a normative cost-sensitive benchmark, meets the requirements for a non-degenerate and interpretable behavioral environment.

Several limitations are acknowledged. The participant sample will consist of non-experts, limiting external validity to professional decision-making contexts; however, this is consistent with the majority of prior work in this area. The within-subject design introduces potential carryover effects, which are partially mitigated by the Latin square counterbalancing but will be examined via block-order analyses. The 18-case design, while sufficient for main-effect detection, provides limited power for three-way interactions, which are treated as exploratory. The predicted probability range in the selected cases (0.055–0.423) reflects the realistic distribution of credit risk model outputs but may limit generalizability to domains where AI systems produce more extreme probability estimates.

Data collection targeting $N = 100$ participants is planned for immediate deployment following the completion of the web-based experiment platform. Future work will report the full experimental results and extend the analysis to examine the non-linear relationship between model correctness and reliance across uncertainty levels.

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APPENDIX A

FORMAL METRIC DEFINITIONS

Metric–Hypothesis Alignment. Metrics are selected to correspond directly to the hypotheses. Accuracy and reliance-

based metrics correspond to H1–H4, while calibration-related metrics support H5–H6.

Let $y_i \in \{0, 1\}$ denote the true outcome for case i , $\hat{p}_i^{\text{model}}$ the model-predicted probability, and $\hat{p}_{i,k}^{\text{system}}$ the participant’s final probability under protocol k .

Weight of Advice (human-first only):

$$\text{WOA}_i = \frac{\hat{p}_i^{\text{final}} - \hat{p}_i^{\text{init}}}{\hat{p}_i^{\text{model}} - \hat{p}_i^{\text{init}}} \quad (2)$$

WOA is undefined when $\hat{p}_i^{\text{model}} = \hat{p}_i^{\text{init}}$; such cases are excluded.

Reliance Decomposition. Each AI-assisted trial is classified into one of four mutually exclusive outcomes based on model correctness $M_i = \mathbf{1}\{\hat{y}_i^{\text{model}} = y_i\}$ and whether the participant follows the model $F_i = \mathbf{1}\{\hat{y}_i^{\text{final}} = \hat{y}_i^{\text{model}}\}$:

TABLE IV
RELIANCE OUTCOME CLASSIFICATION

	Follows ($F = 1$)	Overrides ($F = 0$)
Correct ($M = 1$)	Beneficial reliance	Harmful override
Incorrect ($M = 0$)	Over-reliance	Beneficial override

The **Discrimination Ratio** is: $\text{DR} = \frac{\text{Rate}(\text{Beneficial override})}{\text{Rate}(\text{Over-reliance})}$. $\text{DR} > 1$ indicates selective override of incorrect AI, the behavioral signature of appropriate reliance [7].

Expected Calibration Error (fixed equal-width bins):

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)| \quad (3)$$

APPENDIX B MIXED-EFFECTS MODEL

$$\begin{aligned} \text{logit}(P(\text{Correct}_{ijk} = 1)) \sim & \beta_0 + \beta_1 \text{MC}_i + \beta_2 \text{Protocol}_k + \beta_3 \text{Diff}_i \\ & + \beta_4 (\text{MC}_i \times \text{Protocol}_k) \\ & + \beta_5 (\text{MC}_i \times \text{Diff}_i) \\ & + \beta_6 (\text{Protocol}_k \times \text{Diff}_i) \\ & + \beta_7 (\text{MC}_i \times \text{Protocol}_k \times \text{Diff}_i) \\ & + u_j + v_i \end{aligned} \quad (4)$$

where $u_j \sim \mathcal{N}(0, \sigma_u^2)$ is a random intercept for participant j and $v_i \sim \mathcal{N}(0, \sigma_v^2)$ is a random intercept for case i . The three-way interaction β_7 is treated as exploratory.